

Problem Set 4

Intermediate Macroeconomics, EC2211

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Due Tuesday, October 21 at 8 am. Worth a maximum of 4 credit for the final grade. Send one solution per group to your seminar teacher through Athena.

1 Analyzing macroeconomic events with the IS curve (1 p)

(Exercise 3 in Jones chapter 11) Consider the following changes in the macroeconomy. Show how to think about them using the IS curve, and explain how and why [U.S.] GDP is affected in the short run.

- a. The government offers a temporary investment tax credit: for each dollar of investment that firms undertake, they receive a credit that reduces the taxes they pay on corporate income.
- b. A booming economy in Europe this year leads to an unexpected increase in the demand by European consumers for U.S. goods.
- c. U.S. consumers develop an infatuation with all things made in New Zealand and sharply increase their imports from that country.
- d. A housing bubble bursts so that housing prices fall by 20% and new home sales drop sharply.

2 A consumption boom (1 p)

(Exercises 3 and 10 in Jones chapter 12) Using the IS/MP diagram, explain what happens to the economy if there is a temporary consumption boom that lasts for one period.

- a. Initially, suppose that the central bank keeps the nominal interest rate unchanged.
- b. Suppose you are appointed to chair the Federal Reserve. What monetary policy action would you take in this case and why? Refer to the IS/MP diagram.
- c. Now explain what happens in the full short-run model (including the Phillips curve and allowing the economy to evolve over time). Do this for both parts (a) and (b), and be sure to provide graphs of output and inflation over time.

3 Crowding out (0.5 p)

(Exercise 11 in Jones chapter 13) Consider a simplified version of the Taylor rule, in which monetary policy depends only on the output gap:

$$r_t - \bar{r} = \xi \tilde{Y}_t.$$

- Draw an IS/MP diagram, but instead of the usual MP curve, plot the simplified version of the Taylor rule. You might label this curve MPR for "monetary policy rule."
- Now consider the effect of a positive aggregate demand shock in the IS/MPR diagram. (An example might be a fiscal stimulus.) Compare and contrast the effect of this shock on the economy in the standard IS/MP diagram versus the IS/MPR diagram. Why is the result different?
- Economists refer to the result in the IS/MPR diagram as "crowding out". What gets crowded out and why?

4 A financial crisis (0.5 p)

(Exercise 1 in Jones chapter 14) Suppose the economy starts with GDP at potential, the real interest rate and the marginal product of capital both equal to 3%, and a stable inflation rate of 2%. A mild financial crisis hits that raises the financial friction from zero to 2% (in other words, for any level of the central bank's nominal policy rate, the real interest rate increases by 2 percentage points).

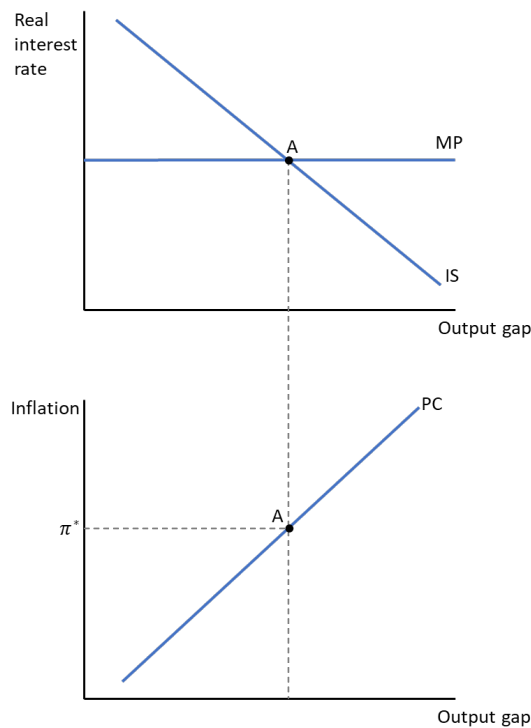
- Analyze the effect of this shock in an IS/MP diagram.
- What policy response would you recommend to the Federal Reserve? What would be the effect of this policy response on the economy?
- How would your answer to part (b) change if the financial crisis were very severe, raising the financial friction to 6%?
- What other policy responses might be considered in this case?

5 A shock to inflation expectations (1 p)

Consider an economy that is in points A in the graphs below. Then, in the beginning of period $t = 1$, inflation expectations suddenly rise to $\pi_1^e > \pi^*$.

In all other time periods, inflation expectations are adaptive and determined by the inflation rate in the previous period, $\pi_t^e = \pi_{t-1}$.

- Suppose that the central bank is passive and holds the *nominal* interest rate constant both in period $t = 1$ and $t = 2$. Use the IS-MP-PC framework to illustrate how the economy develops in these two time periods.



- b. Suppose instead that the central bank quickly sees that inflation expectations have increased. It therefore changes its policy rate early in period $t = 1$, and then continues to adjust its policy rate with the objective to stabilize both inflation and output. Use the IS-MP-PC framework to illustrate how the central bank responds and how the economy then develops. Does your answer depend on how the central bank prioritizes inflation stability relative to output stability?
- c. Consider part (b) again, but use the AS-AD framework to illustrate the same scenario.

Recommended extra exercises

Do not hand in solutions to the exercises below! Exercises marked by a * are solved in the textbook. Other exercises may be discussed in your sessions if time allows.

- Jones chapter 11: Exercises *2 and 9
- Jones chapter 12: Exercises 6, *12 and 13
- Jones chapter 13: Exercises 3, 5, 6, 7, 8, *9 and 10
- Jones chapter 14: Exercise *2